

IMO Syllabus

The syllabus for International Mathematics Olympiads is pre-degree college mathematics. The areas covered are:

Elementary Number theory:	Peano's Axioms, Algebraic properties of $\mathbb{N}$, order properties of $\mathbb{N}$, principle of mathematical induction, First principle of induction(FPI), second principle of induction (SPI), third principle of induction, basis representation theorem, integers, properties of integers, the greatest integer function, divisibility, tests of divisibility, greatest common divisor of two integers, Euclid's algorithm, the unique factorization theorem, congruences, chinese remainder theorem, more on divisibility, number of divisors of a composite number, number of ways in which a composite number can be resolved into two factors, Number of ways in which a composite number can be resolved into two factors which are prime to each other, Sum of divisors of a number, the highest power of a prime which is contained in $n!$, Euler's Totient function, Divisibility of the product of k consecutive integer by $k!$, theorems of Fermat and Wilson, converse of Wilson's theorem, solution of equations in integers.
Inequalities	Introduction, elementary inequalities, absolute value, inequality of the means, Cauchy-Schwarz inequality, Tchebychef's inequality.
Equations	Introduction, Polynomial functions, division remainder theorem, factor theorem, converse of factor theorem, Polynomial equations, relation between the roots and coefficients of a polynomial equation, symmetric form of an equation, common roots of polynomial equations.
Combinatorics	Introduction, Use of three diagrams, rules of sum and product, some applications of the rule of the sum, principal of inclusion and exclusion, An interesting case of principle of inclusion and exclusion, pigeon hole principle, recurrence relations, binomial co-efficients, proof of binomial theorem for a positive integral index.
Some of special topics	Introduction, clocks, identities, functional equations.
Geometry	Introduction, straight lines and triangles, Ceva's theorem, Menalaus theorem, construction of triangles, parallelograms and rhombuses, circles, circum-circles, incircle, excircles, nine-point circle, Simson's line, Common tangent to two circles, centers of similitude of two circles, area of a quadrilateral, ratio and proportion, Lehmus Steiner theorem, Ptolemy's theorem.

The syllabus does not include calculus and statistics.

The typical areas for problems are: Number theory, Geometry, Algebra and combinatorics.

The syllabus is in a sense spread over class IX to class XII levels, but the problems under each topic are of an exceptionally high level in difficulty and sophistication.